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Abstracts

A predevelopment program for the HM60 engine is going on at Volvo Flygmotor AB. One of the items in this program is the development of a supersonic turbine stage. Due to the specified high pressure ratio a design with supersonic blade flow is attractive in the first stage of the two stage LH2-turbine and in the single stage LOX-turbine.

The rotor blade flow has been studied in wind tunnel tests and a first design has been turbine tested. From the initial tests some basic problems were identified and further wind tunnel experiments have given promising solutions. Turbine tests of a redesigned turbine are in preparation. Some interesting observations are:

- The boundary layers have a considerable growth and the blade channels must be properly compensated for the displacement.
- The boundary layer stability is critical on the convex side of the blades and the profile must be properly designed in order to avoid separation.
- The leading edges must be properly designed in order to avoid losses and shock waves disturbing the nozzles.

By testing the turbines with hot compressed air instead of the specified H_2-mixture the development tests can be performed at realistic condition at much lower cost.

1. Introduction

In order to improve the performance of the ARIANE Launchers family, and to reduce specific cost of payloads in LEO and GEO, CNES (Centre National d'Etudes Spatiales) has conducted trade off studies which lead to the need for a new LOX-LH2 rocket engine named HM60, generating 900 KN thrust in vacuum. The engine will be developed by Société Européenne de Propulsion in France in cooperation with other European Countries. Trade off studies on the engine cycle showed that, in order to reduce production costs and development risks, the gas generator cycle has been selected. At present the nominal characteristics of the engine are : combustion pressure : 100 bar, mixture ratio 5.1, specific impulse 444 s, single gas generator feeding, in parallel two independent turbopumps.

The HM60 will be developed and produced in cooperation between several European companies. At this time three of them are in progress to design the engine and its major subsystems.

- Société Européenne de Propulsion, France, for the integrated engine, turbopumps and accessories.
- Messerschmitt-Bölkow-Blohm, West Germany, for the thrust chamber, the igniter and the gimbal joints.
- Volvo Flygmotor, Sweden, for the nozzle extension and the turbines.

A predevelopment program is going on at Volvo Flygmotor since 1981 and one of the items in this program is development of a supersonic turbine stage.

2. Turbine specifications and preliminary layout

The HM60 engine has two turbines, one for the LH2-pump and one for the LOX-pump. Their main datas are shown in Fig 1. Special for those turbines are the high pressure ratio and the high available enthalpy in relation to the number of stages and to the blade speed.

Turbine	LH2	LOX
Shaft power nominal KW	9120	2350
" " max "	14870	3380
" speed nominal rpm	38600	14500
" " max "	44800	16200
Inlet pressure nominal kPa	7500	6000
" temperature nominal K	876	876
Pressure ratio	20.5	17.1
Mean diameter mm	201	300
Blade speed nominal m/s	406	228
Number of stages	2	1
Specific heat of the gas J/kg • k	8040	8040

Fig 1 HM60 Turbines main specification

For the LOX-turbine in which a very high pressure ratio has to be used in a single stage a high supersonic inlet Mach number must be used in order to obtain a reasonable efficiency. Due to the small volume flow this turbine must be designed for partial admission.

For the LH2-turbine two alternative layouts have been regarded, one with the main part of the expansion in the first nozzles (velocity compound) and another with the expansion more equally distributed between the two stages (pressure compound). In the first alternative the first rotor will operate at an inlet relative Mach number close to 2 and the second rotor at about 0.7. At such a high inlet Mach number as 2, strong compression shocks must be avoided as they cause high losses and therefore the flow must be kept supersonic through the rotor blade channels.

In the pressure compound alternative the two rotors are operating at a relative inlet Mach number of ~ 1.3. At this level the shock losses are small and the blades can be designed for a normal shock at the leading edge and a mixed super/subsonic flow in the blade channels.

In spite of the shock problems, the velocity compound alternative was chosen for the following reasons.

- The predicted performance was slightly higher.

- The blade heigth was more reasonable than the very short blades required in the first stage of a pressure compound turbine.

- As the LOX-turbine required supersonic blade flow a supersonic first stage in the LH2-turbine will not increase the development cost significantly.

3. Development and test strategy

As in most development programs the tests tend to take the dominant part of the cost and the required time. In order to keep down cost and time it is important to arrange the test program in an efficient way and to take advantage of existing test plants as long as possible. Among our existing resources are a number of supersonic wind tunnels and a long experience from supersonic aerodynamics. We also have a big air plant with capacity to deliver large quantities of compressed air. This plant is equipped with heaters and with ejectors to keep down the outlet pressure. Therefore the following test program was chosen.

- The supersonic blade flow was to be studied and the profiles optimized by wind tunnel tests. For those tests a special test section was designed and a wind tunnel rebuilt to feature the tests.

- The turbine development tests as well as the delivery acceptance tests are to be run with the turbines fed by hot compressed air.

- Tests with the specified high pressure hydrogen mixture will be run only as tests of the complete turbopump units, performed under the responsibility of SEP.

- As it was regarded as important to obtain turbine test experiences in an early stage of the program an experimental turbine with exchangeable elements was built. For the initial tests it was equipped with elements of a preliminary design which will later on be replaced by gradually more refined ones.

Tests with the turbines fed by hot compressed air will simulate the specified conditions very well as:

- The turbines can be tested in full geometric scale.

- The turbines can be tested at a correct pressure ratio. The specific heat ratio is almost the same for the hot air as for the specified gas and thus the turbines are tested at a correct Mach number similarity.

- The turbines can be tested at a correct relation between the gas- and the blade-speed and thus the velocity triangles will be similar.

- The Reynolds number will be of a correct order of magnitude.

4. Design of a first generation supersonic nozzle- and blade-profiles

The nozzles were designed for a pressure ratio of ~ 20, corresponding to an outlet Mach number of 2.6. As it was desired to keep the nozzles as short as possible a design with a sharp corner throat section, according to the principles given in ref [1], was chosen. The shape of the nozzles was computed by a program for two dimensional isentropic flow according to the method of characteristics and the curved walls where adjusted due to the boundary layer displacement.

The basically two-dimensional eighteen nozzles where wrapped around to a cylindrical ring, designed for manufacturing by a radially fed EDM-electrode and contained in an outer cylindrical ring.

The rotor blades where designed for the conditions in the central part of the nozzle jets at the mean diameter and the blade cross section was kept constant over the entire blade height. The design condition was:

- Relative inlet Mach number 2.25
- Relative inlet flow angle 21°
- Static pressure ratio 1:1

The design of the blade profiles was based on the theory given in refs [2] and [3] and for the calculations a two-dimensional isentropic computer program according to the method of characteristics was used. The prescribed surface Mach number distribution is shown in Fig 2. As from the mechanical point of view, the cord ought to be as short as possible, the first design used a Mach number gradient as high as could be permitted by the criteria given in ref [2]. From the theoretically computed profiles with infinitely thin leading and trailing edges the following corrections where made.

- As compensation for the boundary layer displacement the height of the blades was increased by 20 % from the inlet to the outlet.
- In order to get a thickness of the edges sufficient from the mechanical point of view the concave and the convex sides of the theoretical profile were moved away and the pitch correspondingly increased.

The nozzle- and blade-profiles are shown in Fig 3.

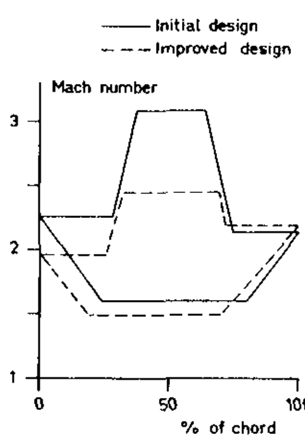

Fig 2 Theoretical surface Mach number distribution

5. Wind tunnel test of the first generation rotor blades

The wind tunnel is shown in Fig 4. Downstream the Laval nozzle with its obliquely cut extension is a cascade of four profile models in the scale 2:1. The central blade channel is used for the aerodynamic observations and measurements and the two side channels are intended to create realistic boundary conditions at the leading and the trailing edges. The outlet is collected into three plenum chambers, each connected to an ejector via a separately controlled line. The central flow plenum is controlled for a correct blade pressure ratio and the other two are controlling the nozzle spill over flows. The tunnel has two alternative sets of side walls, one with diverging walls simulating the increasing blade height and another with parallel walls simulating the computed two-dimensional flow.

The flow conditions around the leading edges and in the blade channel were studied by Schlieren optical observations and by pressure measurements. As a complement the wet paint method was used for study of boundary layer separations. From the wind tunnel tests of the first generations rotor blades the following conclusions can be drawn:

- There was a boundary layer separation on the convex side of the blades. The separation starts at the central section of the blade at a distance from the leading edge corresponding to ~25 % of the chord. This indicates that the criteria given in ref [2] may be a little too optimistic.
- There was a considerable boundary layer growth on the convex side of the blade profiles, requiring a compensation of the channel width.
- The leading edges must have a wedge angle low enough to avoid shocks giving pressure losses and risk for disturbances at the nozzle outlet.
- At both ends of the blade channels there is a considerable overflow of gas from the concave to the convex side. This overflow is expected to have the following consequences:
  - A very thin boundary layer and a very little risk for separation on the concave side.
  - Increased boundary layer thickness and thus increased risk for separation on the convex side.
  - Excessive losses if very low aspect ratio is used.

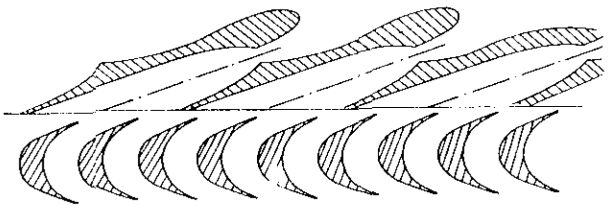

Fig 3 First generation nozzle- and blade-profiles

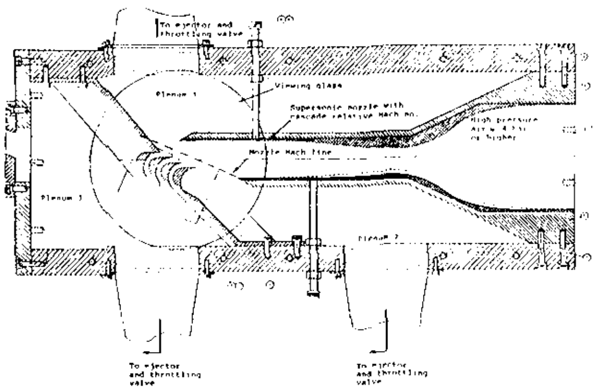

Fig 4 Supersonic blade cascade wind tunnel

6. Turbine tests of a first generation supersonic stage

The experimental turbine is shown in Fig 5. In this test it was equipped with nozzles and blades of the first generation described above. The mean blade diameter was 200 mm and the height of the nozzles was 16 mm. The blades were unshrouded and had a tip clearance of 0.35 mm.

The turbine was fed by air of 8.4 bar heated to 500 K. By this heating a condensation of moisture is prevented. The outlet was connected to an ejector making it possible to test the turbine over a wide range of pressure ratios. The turbine shaft was connected to a water brake from which the torque and the speed was measured. The static pressures before and after the rotor was determined as the averages from measurements on the outer and inner diameter of the duct.

Unfortunately the test results indicate that the flow was choked at the rear end of the blade channels at stage pressure ratios above ~ 10. At the choked condition the nozzle pressure ratio was only ~ 6.6 instead of the design pressure ratio of ~ 20.

Due to the choking, the nozzles as well as the rotor blades, where operating at conditions very different from what they were designed for. In spite of that their losses were fairly low. From the measured torque and pressures the velocity coefficients (real average outlet velocity in relation to the isentropic) was calculated to be ~ 0.88 for the nozzles and 0.80 for the rotor blades.

One interesting observation was the big differences between the static pressures measured at the outer and inner radius of the duct. Although the ratio of the radii is only ~ 1.17 the measured pressure ratio was as high as ~ 1.6. Calculations confirm the measured ratio and they indicate that the pressure ratio would have been much higher if the nozzles had been operating at their design conditions. Due to the differences in pressure and the corresponding differences in Mach number and flow direction a rotor designed for the mean condition must operate mis-matched at its top- and root section. Those differences are a consequence of the "wrapped around" nozzle design and they show the importance of a redesign to straight nozzles.

Due to a delay in the wind tunnel program the experimental turbine had to be frozen before the completion of the wind tunnel test. Later wind tunnel tests explain fairly well the turbine choking. The blades have an insufficient compensation for the boundary layer displacement which causes an increase of the static pressure and a decrease of the Mach number. The blades also have an insufficient boundary layer stability and in combination with the increased back pressure it leads to separation. The separation causes an increased back pressure of the nozzles which starts a chain reaction in which the mismatched nozzles gives mis-matched inlet conditions to the rotor and so on until the flow is completely choked.

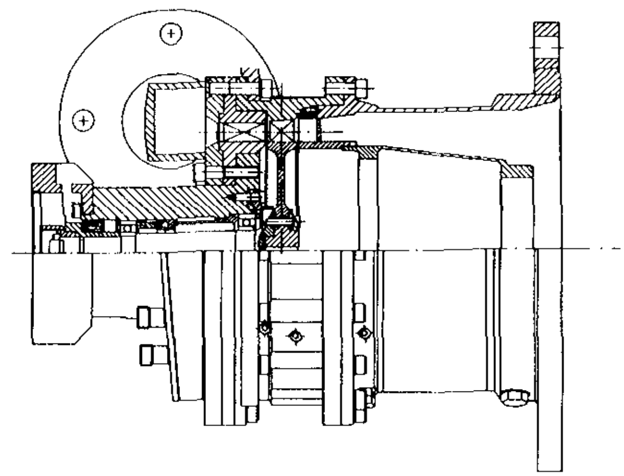

Fig 5 Experimental turbine

7. Development of improved supersonic rotor blades

The initial wind tunnel and turbine tests showed that efficient supersonic blades must satisfy the following requirements.

- A leading edge geometry giving a shock pattern without considerable losses and without serious interaction on the nozzle outlet.

- A stable boundary layer on the convex side of the blades.

- A blade channel area distribution, properly compensated for the flow losses and the boundary layer displacement.

In the development of the improved rotor blades each of those problem areas have been studied in wind tunnel experiments. In order to avoid additional losses and shock vaves that may disturb the nozzles, the leading edge wedge angle ought to be kept below 15 deg. Regarding the concave side boundary layer no separation was observed at blades designed for the lower Mach number gradients shown in Fig 2. The losses were found to be decreased if the turbulence transition was fixed by a row of very small steel balls close to the leading edge. This observation indicates a late natural transition and that the boundary layer stability can be affected of the leading edge design. The concave side was found to have a very thin boundary layer without any tendency to separation. Regarding the channel areas a method of determinating the required correction of the theoretical area distribution has been tested and found to be sufficient.

The result of the development work is a redesigned blade profile with, according to the wind tunnel results, promising properties.

8. Turbine test of a second generation supersonic stage

The nozzle ring and the rotor blades for the experimental turbine have been redesigned and the hardware for a turbine test is in preparation. The pressure ratio and the inlet Mach number are slightly decreased due to revised layout of the second stage and the new first stage is designed for a relative inlet Mach number of 2.0 instead of 2.25. In order to obtain a more even Mach number distribution the "wrap around" nozzle ring has been replaced by a ring of straight center line two-dimensional nozzles. The nozzles will be produced by EDM in two steps, the supersonic part by an electrode fed from the outlet side in the direction of the nozzle centre and the subsonic part by an electrode fed from the inlet side. The two parts produced by EDM are connected by a radially drilled cylindrical bore. The nozzles have a direction giving jets that are tangential to the inlet of the rotor blades. The rotor has blade profiles according to the wind tunnel tests reported above. They have a constant cross section and will for the experimental turbine be produced by a radially fed EDM electrode and the rotor will be equipped with a shrink fitted shroud ring. For the future production turbines the blades are intended to be separately forged and fir-tree attached to the disk. The new nozzle and blade design are shown in Fig 6.

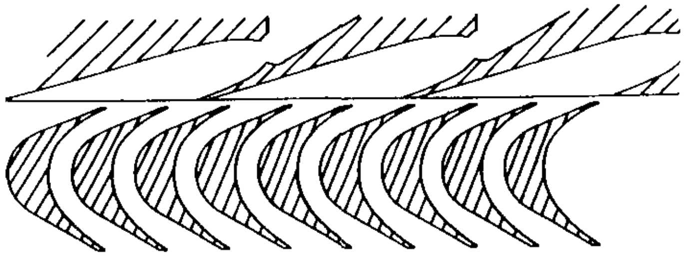

Fig 6 Improved nozzle and blade profiles

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